BRANCHES AND GEOMETRIC BRANCHES AT A POINT ARE COUNTED BY THE FIBRES OF THE NORMALIZATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Stalk of the normalization as an integral closure. Stacks Project, Morphisms, Lemma morphisms-lemma-stalk-normalization (tag 035Q). Identifies the stalk of the pushforward of the structure sheaf along the normalization with the integral closure of O_{X,x}.
- Integral extensions and the correspondence of primes; algebraic branch count. Stacks Project, Algebra, Lemmas algebra-lemma-integral-no-inclusion (tag 00GS) and algebra-lemma-integral-going-up (tag 00GT); More on Algebra, Lemma more-algebra-lemma-number-of-branches-1 (tag 0C1R). The first two turn the fibre over x into maximal ideals of the integral closure and the third supplies the branch formulas.

Source reference. Original reference 035N: sources/stacks-chow/tags.txt, source lines 4134–4134. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0C3B: sources/stacks-chow/tags.txt, source lines 15186–15186. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00GT: sources/stacks-chow/tags.txt, source lines 619–619. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00GU: sources/stacks-chow/tags.txt, source lines 620–620. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0C37: sources/stacks-chow/tags.txt, source lines 15182–15182. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a scheme. Assume every quasi-compact open of X has finitely many irreducible components. Let $\nu : X^\nu \to X$ be the normalization of X. Let $x \in X$.*

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(1) *The number of branches of X at x is the number of inverse images of x in X^ν.*
(2) *The number of geometric branches of X at x is $\sum_{\nu(x^\nu)=x}[\kappa(x^\nu) : \kappa(x)]_s$.*

Proof. First note that the assumption on X exactly means that the normalization is defined, see Morphisms, Definition 035N. Then the stalk $A' = (\nu_*\mathcal{O}_{X^\nu})_x$ is the integral closure of $A = \mathcal{O}_{X,x}$ in the total ring of fractions of A_{red}, see Morphisms, Lemma 0C3B. Since ν is an integral morphism, we see that the points of X^ν lying over x correspond to the primes of A' lying over the maximal ideal $\mathfrak{m}$ of A. As $A \to A'$ is integral, this is the same thing as the maximal ideals of A' (Algebra, Lemmas 00GT and 00GU). Thus the lemma now follows from its algebraic counterpart: More on Algebra, Lemma 0C37. □